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What task to offload to GenAI in the writing feedback process? – Effects of task-offloading approaches on EFL learners' writing skill development

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ABSTRACT

Generative artificial intelligence (GenAI) brings mixed blessings to learning, and it is important to figure out ways to optimize the integration of GenAI that augments human intelligence. This study explored different approaches to offloading tasks to GenAI in the writing feedback process and the impact on student writing development. Adopting a quasi-experimental design, this study compared 101 university English as a foreign language (EFL) students' writing skill development under three GenAI task-offloading conditions: (1) GenAI-draft, learner-evaluate, and learner-revise; (2) learner-draft, GenAI-evaluate, and learner-revise; and (3) learner-draft, learner-evaluate, and learner-revise. ANCOVA analyses revealed that involving GenAI in the feedback process led to greater writing gains, and that positioning GenAI as draft writer and learners as feedback provider carried the most benefit. The findings draw attention to the nuances in task-offloading approaches, and advocate offloading initial production to GenAI and onloading evaluation and revision to learners. The study further revealed that learners in this condition exhibited greater instances of interactive engagement (learning-oriented interaction in particular) with GenAI, performed better in peer review tasks that tested the application of metacognitive knowledge of argumentative writing, and reported higher cognitive engagement, deeper metacognition of argumentative writing and of AI's limitations, as well as greater writing self-efficacy. These psychological processes were influenced by various motivational, cognitive, and metacognitive factors. A conceptual model of the psychological processing under this task offloading condition and AI's role was proposed based on the research findings.

1. Introduction

Generative artificial intelligence (GenAI) brings mixed blessings for writing feedback. On the one hand, it boosts the customization, interactivity, and ubiquity of feedback practices (Ali et al., 2023; Jeon et al., 2023). GenAI feedback is recognized for being (1) impersonal, reducing emotional stress during the feedback process (Zou et al., 2025); (2) elaborative, enhancing learning opportunities (Escalante et al., 2023; Lee & Moore, 2024); and (3) timely, boosting learner engagement and subsequent learning (Alsofyani &

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techniques for different English genres (i.e., expository, narrative, descriptive, and argumentative) taught by the same instructor. The module for the argumentative genre lasted four weeks, during which students received instruction on constructing argument structure, formulating logical claims, utilizing evidence and transitions, and addressing counterclaims. All participants were English majors, with an average age of 20.39 years (SD = .88). Their English proficiency was of intermediate level (CEFR level B2), as evidenced by their scores in the College English Test Band 4 (M = 485.29, SD = 40.58). The participants reported an average of seven months' prior experience with GenAI tools, and their average frequency of GenAI use was between "a few times per week" and "less than 30 min per day". Only 25 students (24.8%) reported using GenAI tools on a daily basis. Appendix A summarizes the demographic characteristics of participants.

Before conducting this study, we obtained ethical approval from the institutional review board of the authors' university and secured informed consent from all participating students.

3.2. Research design

The study employed a quasi-experimental design, assigning the three intact classes into three intervention conditions: Class 1 – the GenAI as Feedback Receiver condition (GenAI-draft, learner-feedback, learner-revise); Class 2 – the GenAI as Feedback Provider condition (learner-draft, GenAI-feedback, learner-revise); Class 3 - the No GenAI Involvement condition (learner-draft, learner-feedback, learner-revise). Participants completed four out-of-class writing tasks using a structured worksheet with three phases – drafting, evaluation, and revision. Fig. 1 illustrates the specific procedure across the three conditions.

3.3. Research procedures

Fig. 2 shows the research procedure. The study lasted seven weeks. In Week 1 (one week prior to the argumentative writing module), all participants completed a pre-test on argumentative writing, and a pre-intervention questionnaire on prior GenAI experience and writing self-efficacy in class, followed by a training that familiarized the participants with both the argumentative writing assessment rubric and the platform that they would use during the intervention. This rubric, adapted from Nakkaew and

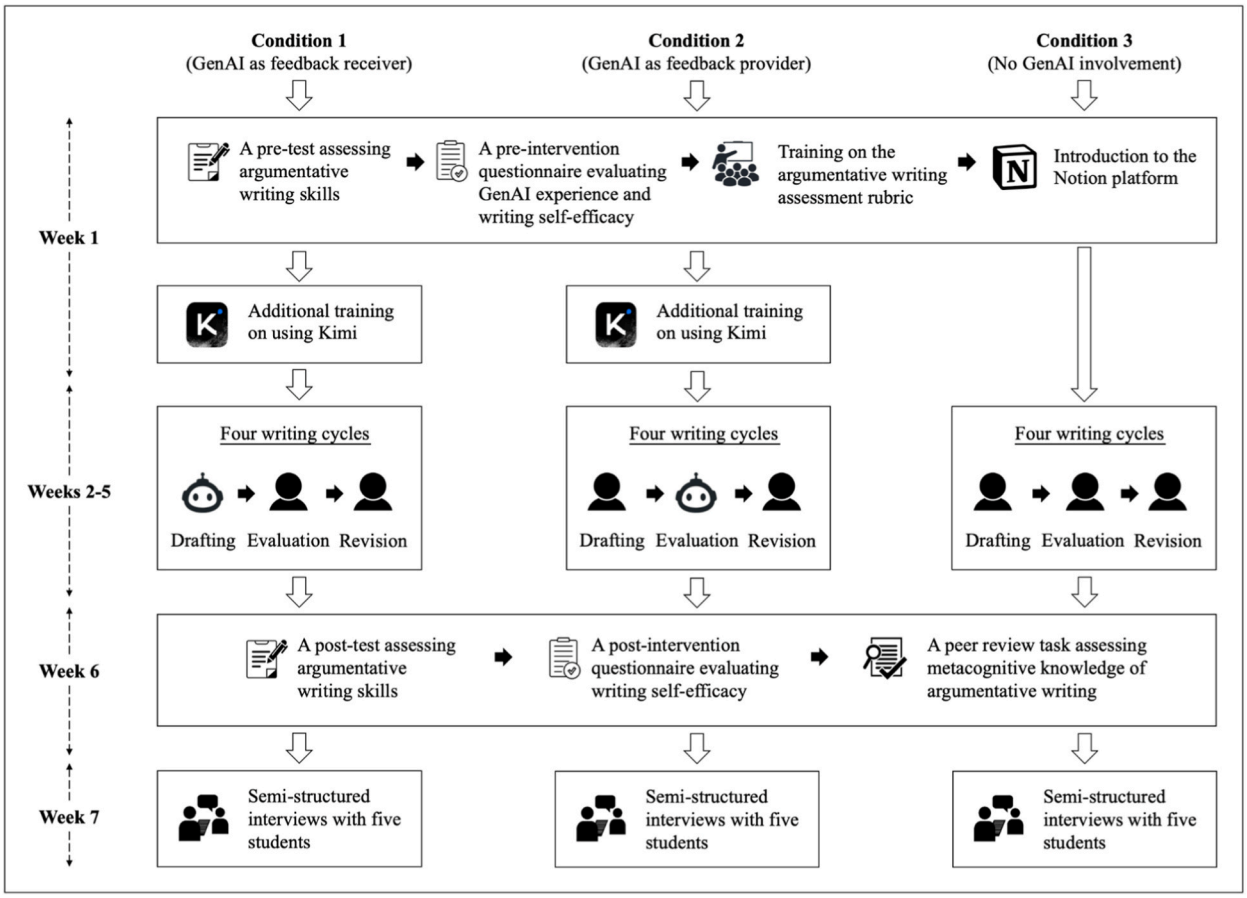

Fig. 2. Research procedures.

Adunyarittigun (2019), assesses six key aspects of argumentative writing: introduction, reason and evidence, counterargument and refutation, conclusion, organization and transitions, and language use (see Appendix B). Each criterion in the rubric was explained to the participants. To enhance active engagement with the rubric, an evaluation worksheet, containing a checklist for each criterion and the justification for evaluation (see Appendix C), was developed for students in Conditions 1 and 3. Participants were also introduced to the Notion platform, a collaborative workspace that creates a structured environment for individual writing tasks and allows the integration of external tools (Osawa, 2023), chosen to facilitate students' workflow and streamline data collection (see Supplementary Materials for the Notion page for each condition). Participants in Conditions 1 and 2 received additional training on using Kimi, a free GenAI tool developed by Moonshot AI, chosen for its validated abilities to generate coherent, contextually relevant text while providing high-quality writing feedback (Jiang, 2025; Yao et al., 2025).

From Week 2 to Week 5 (during the argumentative writing module), participants completed four out-of-class argumentative writing tasks on the Notion platform, one task each week. The writing tasks were adapted from the GRE's 'Analyze an Issue' task (see https://www.ets.org/gre/revised_general/prepare/analytical_writing/issue/), carefully selected to elicit opinions on educational issues from different perspectives: parent/employer (praise positive actions and ignore negative ones), government (free education for all), society (modern luxuries on character development), and university (interest-driven vs. job-oriented college majors) (see Appendix D for the writing prompts). To encourage students to take the writing practice tasks seriously, these writing assignments were incorporated into the course's participation score, with no teacher feedback nor in-class discussions on the writing products. Participants in Condition 3 were instructed not to use any GenAI tools for writing assistance, and the instructor periodically checked with the students and conducted qualitative reviews of their writing samples to identify any inconsistencies in writing style that might suggest undeclared GenAI assistance.

After the completion of the argumentative writing module, in Week 6, a post-test on argumentative writing skills and a post-intervention questionnaire on writing self-efficacy were administered. In addition, all participants completed a peer review task.

In Week 7, individual semi-structured interviews were conducted with five students from each condition.

3.4. Data sources

3.4.1. Pre-and post-test on argumentative writing skills

Participants' argumentative writing performance was assessed via a 40-min in-class pre-test and post-test, with no access to GenAI or any reference materials. The tasks were also selected from the GRE's 'Analyze an Issue' assessment, deliberately focusing on education topics from the learner perspective (pre-test: long-term goals vs. instant fame in life; post-test: active questioning vs. passive acceptance of teaching) to enhance the prompt comparability in genre, familiarity, difficulty, and rhetorical demands, and also differentiate them from the prompts in the practice sessions (see Appendix D). Two experienced writing instructors vetted the prompts to ensure the equivalence in cognitive complexity, linguistic difficulty, and rhetorical requirements. GRE's 'Analyze an Issue" assessment is a measure of analytical writing skills, which engages students to develop and support a position on the issue featured in the prompt, requesting the formulation and expression of logically connected ideas supported with applicable examples. The construct validity of this measure has been substantiated in previous research (Breyer et al., 2014).

3.4.2. Pre- and post-questionnaire on writing self-efficacy

We assessed students' writing self-efficacy using seven researcher-developed, six-point Likert scale (1 = strongly disagree to 6 = strongly agree) items on the ideation and convention aspects of argumentative writing (Bruning et al., 2013). The items were reviewed by a writing research expert and pilot-tested with another group of undergraduate students to ensure clarity and reliability. The items (see Appendix E) demonstrated strong internal consistency in both pre- ($\alpha = .88$) and post-questionnaire ($\alpha = .91$).

3.4.3. Peer review task

Given that metacognitive knowledge of writing, being a conceptual element of feedback literacy (Teng & Ma, 2024), is positively associated with the quality of peer review (Bui & Kong, 2019), and that performance metrics are advocated as concrete ways to capture metacognitive control (Zhang et al., 2024), we assessed students' peer-review performance, as a behavioral proxy of metacognitive knowledge of argumentative writing, on Week 6, where they were instructed, in 25 min, to offer feedback on a peer essay on the same topic written by a student from a previous cohort (see Appendix F).

3.4.4. GenAI conversation logs

We collected conversation logs with Kimi during the four writing tasks from students in Conditions 1 and 2.

3.4.5. Semi-structured interviews

Fifteen students were invited to participate in individual semi-structured interviews in Chinese, their native language, each lasting approximately 30 min. To capture a diverse range of perspectives, participants were selected using maximum variation sampling within each condition to reflect varied writing performance. The interviews elicited students' overall perceptions of their experience, and perceived impact, if any, on writing and self-efficacy and reasons thereof. Students from Conditions 1 and 2 were also invited to share views on the role of GenAI in the process (see Appendix G for the protocol). The interview protocol was pilot-tested with three students prior to the main interviews, leading to minor language refinements to enhance the clarity.

3.5. Data analysis

To answer RQ1 on writing skill development, we evaluated the students' pre- and post-writing performance using the same adapted six-aspect assessment rubric (Nakkaew & Adunyarittigun, 2019) employed during the intervention (see Appendix B). Each essay was scored on a scale from 1 to 4 on each aspect, resulting in a maximum score of 24 for each essay. To ensure scoring reliability, the second and third authors studied the rubric together and then independently coded 25% of the essays, with satisfactory inter-rater reliability: introduction ($\kappa = .85$), reason and evidence ($\kappa = .92$), organization and transitions ($\kappa = .88$), counterclaim and rebuttal ($\kappa = .91$), conclusion ($\kappa = .87$), and language use ($\kappa = .90$). All discrepancies were resolved through discussion. The remaining essays were then scored independently by the second author. To further validate scoring consistency, a random subset of these independently scored essays underwent a secondary review by the third author, with any remaining disagreements resolved through joint arbitration. ANCOVA analysis was then conducted on the post-test score, using pre-test scores as the covariate.

RQ2 examined the three psychological processes. To examine cognitive engagement, we measured both the qualitative dimension – nature of cognitive processes attended to via interview responses, and quantitative dimension – interactive engagement with GenAI via GenAI conversation logs (Alam & Mohanty, 2024). GenAI conversation logs provided the behavioral indicators of interactive engagement, and interview responses tapped further into learner voices on the nature of cognitive processes. We coded GenAI conversation logs on the total number of GenAI prompts, indexing attempts of interactive engagement (Chi and Wylie, 2014), and the number of learning-oriented prompts, indexing the depth of interactive engagement (Oliveira et al., 2025). Learning-oriented prompts were operationalized as prompts that seek guidance and advice from GenAI for learning purposes (see Appendix H for the coding scheme). Following the same reliability validation procedure, the prompts were coded by the second and third authors, and a reliability score of .93 was obtained for the coding, with all discrepancies resolved through discussion. The frequencies were compared between Conditions 1 and 2 using Mann-Whitney *U* test due to the violation of the normality assumption. Thematic analysis was conducted on the interview responses to shed light on the depth of cognitive processing attended to and the level of cognitive engagement (see Appendix J for the coding scheme, which was derived inductively from the data).

To examine metacognition, we assessed the participants' metacognitive knowledge of argumentative writing and of AI (Brown, 1987; Teng, 2025a) through both comparing students' peer review performance to examine learners' ability to apply metacognitive knowledge of argumentative writing and analyzing interview responses to shed light on whether and how the experience impacted both metacognition of argumentative writing and metacognition of AI as well as the influencing factors. The quality of students' peer feedback was assessed using three key features from Kerman et al.'s (2024) evaluation framework that target metacognitive elements: (1) identification (the identification and localization of issues in the essay), (2) justification (elaborations and justifications of the identified issues), and (3) constructive (the inclusion of recommendations and action plans for further improvements). See Appendix I for the coding scheme. Each feedback comment (a full sentence) received a score for each feature, resulting in a maximum score of 6 for each comment. The scores were then aggregated to provide an overall feedback quality score for each student. The coding was done by the second and third authors, following the same reliability validation procedure as above. The inter-rater reliability scores for the feedback quality coding were as follows: identification ($\kappa = .85$), justification ($\kappa = .82$), and constructiveness ($\kappa = .88$). MANOVA analysis and subsequent ANOVA analyses were conducted on the three dimensions. Thematic analysis on the interview responses shed further light on metacognitive knowledge of argumentative writing and metacognition of AI interaction (see Appendix J for the coding scheme).

We examined writing self-efficacy by comparing survey responses for self-efficacy gains and analyzing interview responses to shed light on how the experience impacted or failed to impact self-efficacy as well as the influencing factors. Composite scores were calculated for self-efficacy based on pre- and post-questionnaire responses. ANCOVA analysis was conducted on the post-intervention writing self-efficacy scores, controlling for pre-intervention scores.

Thematic analysis was conducted on the interview data through an evolving process of coding and recoding (Saldana, 2021) to generate codes on the psychological processes in terms of the nature of cognitive engagement (Alam & Mohanty, 2024), metacognition of argumentative writing and of AI interaction (Brown, 1987; Teng, 2025a), and writing efficacy, as well as codes on the motivational, cognitive, and metacognitive barriers to learning in the process (Fiorella, 2023). Based on the interview responses and relevant literature, we conducted structural coding to generate data segments of interest (e.g., segments on AI feedback). Each data segment was then coded using in-vivo or summary codes. In the second-cycle analysis, we first assessed the utility of the codes, and recoded, categorized or discarded codes through cross-participant comparison. We then conducted pattern coding to group codes with similar meaning under analytical categories. For instance, codes like "lengthy responses", "rather tactful and indirect", and "looking very similar" were grouped into the analytic category "limitations of AI responses". Referring to relevant literature, provisional analytic categories were further categorized into theoretical themes such as "metacognition of AI feedback". Theoretical categories that co-occurred were coded into provisional thematic categories and compared across cases to reveal potential relationships (e.g., "metacognition of AI feedback formed motivational barriers to learning"). The interview coding was done by the first and second authors. They first coded one third of the data together to finalize the coding scheme, and then coded the rest independently. Inter-rater reliability, calculated using Cohen's kappa, was .88, with all discrepancies resolved through discussion. See Appendix J for the coding framework.

4. Findings

ANOVA analyses of pre-test scores ($F = .18$, $p > .05$), prior GenAI experience ($F = .25$, $p = .78$), and frequency of GenAI use ($F = .83$, $p = .44$) all showed no significant differences, indicating comparable initial writing abilities and AI experience across the three

groups.

4.1. Writing skill development across the three groups

ANCOVA analysis revealed a significant difference across the three conditions on post-writing test performance: F (2, 97) = 26.63, $p < .001$, partial $\eta^2 = .35$. Post hoc comparisons using Bonferroni correction revealed that both Condition 1 (M = 18.21, SD = 2.21) and Condition 2 (M = 16.30, SD = 1.70) performed significantly better than Condition 3 (M = 15.15, $p = 1.79$), indicating that incorporating GenAI into the feedback process did lead to greater learning (see Table 1).

Moreover, Condition 1 (M = 18.21, SD = 2.21) performed significantly better than Condition 2 (M = 16.30, SD = 1.70), indicating that offloading the drafting task to GenAI for learners to evaluate and revise led to greater learning than offloading the feedback task to GenAI.

4.2. Psychological processes across the three groups

4.2.1. Differences in cognitive engagement

Mann-Whitney U tests on both the total number of GenAI prompts (U = 195; $p < .001$) and of learning-oriented prompts (U = 398.5, $p < .05$) were statistically significant (Table 2). Students in Condition 1 not only interacted significantly more with GenAI but also inputted more prompts for clarification, guidance and advice from GenAI (Median = 20.50; IQR = 22 and Median = 10.50; IQR = 16 respectively) than their counterparts in Condition 2 (Median = 9.00; IQR = 8 and Median = 8.00; IQR = 8 respectively).

The interview responses further revealed that the three groups manifested different levels of cognitive engagement with the assessment rubric and AI output. Participants from the GenAI as Feedback Receiver group (anonymized as P01-P05) reported interacting with AI-generated model text with a learning attitude, as P03 remarked: "My first reaction was to acquire and curate materials for writing. They perceived AI essays of high writing quality, which enhanced their willingness to engage in-depth with the texts (e.g., "AI-generated essays were at a higher level than mine, and I was more willing to study" – P04). Perceiving AI essays as "free of grammatical errors," they reported being able to pay more attention to aspects like "argumentation techniques" (P02), "sentence structure variation" (P04), and "style of argumentation" (P03). They further commented that revising these essays pushed them to study the texts carefully (P02) and engage "thoroughly" and read in greater detail with a critical eye (P03). Evaluating AI essays in reference to the assessment rubric boosted greater reflexivity towards the weaker aspects in their own writing (e.g., "When assessing the points, you would think of your previous essays – what needs attention and what needs to be improved" – P04). Their cognitive engagement was further reinforced by their metacognitive positioning of GenAI as a learning guide (e.g., "guide me on how to write a claim or organize arguments based on my idea" – P02; "It provides me with guidance. I can learn from it." – P04; "Anything I am not sure about, and it will explain for me." – P05), and a learning partner ("It is a learning partner who is at a higher level than me." – P03). These learning-oriented positionings of AI made them engage deeply with AI output: All but one reported prompting AI to evaluate its generated essay in reference to each criterion of the rubric, with the aim to "engage AI to explain whether and how its essay achieved each criterion" and they themselves "judge the rationale provided" (P02). P01 attributed this interaction style to her prior knowledge: "I know what a hook is. So I did not ask it to explain. But instead, I asked it to evaluate whether its essay contained a hook to see whether I agree with its explanation." They reported further interaction with AI (e.g., "challenge AI to elaborate" – P01; "question it iteratively" – P03) to seek elaborations on unfamiliar concepts in the rubric, discuss potential evidence to support a claim, and ask for more sophisticated sentence structures. They further reported deliberate learning engagement with AI output ("I would remember and use the vocabularies that frequently appeared in the texts" – P02).

In contrast, participants from the GenAI as Feedback Provider condition (anonymized as P06-P10) largely reported prompting GenAI to directly revise the essay for them, despite being instructed to revise the essays on their own. Some recounted rephrasing sophisticated language that did not reflect their level to avoid being spotted as plagiarism (P09). They attributed this performance-oriented engagement with AI to cognitive barriers. For instance, P08 recounted lacking the language to make revisions on her own: "Limited language proficiency could not support me to make the revisions despite knowing what needed to be revised." Similarly, P10 reported "asking AI to directly revise the essay for [him]," and, when prompted why, he answered: "I don't know how to revise." Yet, although also asking AI to generate revision samples, P07 took a learning orientation: "I asked it to provide examples that met the criteria so that I could refer to when expanding my essay." She further reported engaging with the part of AI feedback that sounded persuasive by "seeking relevant information on Baidu [a popular internet browser in China]." This learning-oriented engagement was

Table 1
Differences in writing skill development.

	Pre-writing score Mean (SD)	Post-writing score Mean (SD)	Pairwise comparison
Condition 1: GenAI as Feedback Receiver (n = 34)	14.09 (1.66)	18.21 (2.21)	1 > 2 *** 1 > 3 ***
Condition 2: GenAI as Feedback Provider (n = 33)	14.15 (1.60)	16.30 (1.70)	2 > 3 *
Condition 3: No GenAI Involvement (n = 34)	13.91 (1.80)	15.15 (1.79)	

Note. ***$p < .001$; *$p < .05$.

Table 2
Differences in student interaction with GenAI.

		Median (IQR)	Mann-Whitney *U* Test
Total amount of student prompts to GenAI	Condition 1: GenAI as Feedback Receiver (n = 34)	20.50 (22)	195.00***
	Condition 2: GenAI as Feedback Provider (n = 33)	9.00 (8)	
Learning-oriented prompts	Condition 1: GenAI as Feedback Receiver (n = 34)	10.50 (16)	398.50*
	Condition 2: GenAI as Feedback Provider (n = 33)	8.00 (8)	

Note. ***$p < .01$; *$p < .05$.

shaped by her metacognitive positioning of AI as a "strategic advisor" who "broadens perspectives" and provides suggestions and samples for her judgement, unlike the others in this group who tended to view AI as a proxy tool (e.g., "edit my language" – P06; "point out my weak points and improve them for me" – P10). Thus, cognitive barriers interplayed with metacognitive positioning of AI to shape this group's cognitive engagement with the assessment rubric and AI output.

All participants in the No GenAI Involvement group (anonymized as P11-P15) also reported that engaging with the assessment rubric for self-evaluation heightened their self-reflection. But they remarked lacking specific guidance on each criterion (e.g., "The rubric only provided a template without explicit instructions on how to realize each criterion" – P12). Consequently, they reported not being able to engage with some criteria in depth due to the lack of "detailed guidance": P13 expressed the difficulty of "judging whether a certain aspect needs further improvement"; and P14 felt that the rubric "only provided him with a direction," and his ability to make revision was constrained by his "existing knowledge and vocabulary size".

4.2.2. Differences in metacognition

One-way MANOVA analysis revealed a significant difference in the quality of peer feedback across the three conditions (Wilks' $\Lambda = .34$, $F(2,98) = 13.46$, $p < .001$, partial $\eta^2 = .42$), with a significant group effect in all three dimensions: problem identification ($F = 24.15$, $p < .001$; partial $\eta^2 = .33$), problem justification ($F = 17.40$, $p < .001$; partial $\eta^2 = .26$), and constructive advice ($F = 23.70$, $p < .001$; partial $\eta^2 = .33$). The findings suggest that participants across the three conditions varied in their abilities to apply metacognitive knowledge of argumentative writing to assess peer essays. The interview responses further revealed differences in the depth of metacognitive knowledge. All the interview participants, regardless of conditions, reported increased declarative metacognitive knowledge about key elements of argumentative essay writing as a result of the assessment rubric, which, to them, provided "detailed" and "logical" organizational structure that served as a "guiding framework" (P06) for subsequent writing and provided them with "clues on where to start" (P12; P15) and ways to "make the essay more comprehensive and logical" (P02). Yet, participants across the conditions reported varying depths of procedural and conditional metacognitive knowledge. The No GenAI Involvement group reported metacognitive knowledge at the principle level (e.g., "use hooks that draw attention" – P11) with a level of uncertainty, using terms like "somewhat unclear" (P11) and "fuzzy" (P13), in specific techniques. Consequently, they regarded the acquired metacognitive knowledge as a "reminder" of things they "didn't notice before" (P11) and a "general direction" (P15). In contrast, participants in the two GenAI conditions reported greater details in metacognitive knowledge. The GenAI as Feedback Provider group highlighted how the feedback from GenAI drew their attention to key characteristics like diversity – diverse viewpoints and varying evidence – and the importance of cohesive and complex sentence structures. Participants in the GenAI as Feedback Receiver group reported acquiring specific techniques from AI essays, such as transition words (P05) and "discourses of rebuttal" (P02), as well as techniques of varying sentence structures and writing attractive introductions and conclusions (P01, P02, P05). They felt AI essays provided concrete examples (P05), and their sophistication offered greater learning possibilities "not constrained by one's own draft" (P01). They further noted AI's limitations in writing when evaluating AI essays – such as "mechanically piecing together claims without depth" (P03) and "lacking supporting details" (P02) – which helped them derive a "deeper" understanding of key characteristics of a good essay.

Participants also reported various barriers that hindered learning. The barriers frequently brought up by participants from the No GenAI Involvement group concerned values of the assessment rubric. They used words like "simple," "abstract," and "rigid" to describe the rubric and complained about lacking specific guidance on individual elements (e.g., "Each criterion is too general and abstract" – P14). Consequently, P13 interpreted the rubric as "a writing formula" and compared the evaluation-revision experience to "a cloze test". As a result, P12 perceived the whole evaluation-revision experience as providing "great help on structure but less on techniques and thinking". Participants in the GenAI as Feedback Provider group also reported motivational barriers that hindered learning, yet pertaining to the limitations of AI feedback, such as "lengthy responses" (P08), "repetitive" (P06), and "speaking in a rather tactful or indirect manner, skipping some problems in my essays, which made me unaware of these problems for a long time" (P09). These perceived limitations reduced their cognitive engagement with the AI output ("Its revision made my essay very sophisticated, which obviously was beyond my level. I asked it to get rid of that part" – P08), limiting its contribution to the development of metacognitive knowledge. Participants in the GenAI as Feedback Receiver group listed the homogeneity in AI output (e.g., "its essays often followed the same pattern" – P05) as limiting the depth of metacognitive knowledge they could acquire through studying the model texts. Evaluating AI essays made P03 discover problems with their organization and language use, which made her question their value: "It might not be that useful in helping develop deeper understanding." These motivational barriers further interacted with metacognitive

barriers, such as metacognitive attention ("I paid more attention to structure. When it indicated my tone was not strong enough, not formal, I wouldn't pay much attention" – P07), and strategic knowledge of AI prompts ("I can't get AI to understand my intention" – P02; "It couldn't produce the version I wanted, maybe because my prompt was not clear" – P08). Cognitive barriers, like background knowledge, also were salient. Limited language proficiency made P04 take a more "appreciative," rather than critical, stance when analyzing AI model texts. For P11, as the assessment rubric "only raised issues without providing solutions," limited language makes her resort to "fixed vocabularies or sentences" to express a formal tone or "seeking and regurgitating online templates" to create attractive hooks, which she felt made her essay rather "rigid". Understanding of the assessment rubric was also listed as an influencing factor. Most participants from the No GenAI Involvement group reported feeling "bewildered" and disoriented at initial exposures to the rubric due to unfamiliarity with some key terms, and the high cognitive load made some "give up paying attention to all the details in the framework" (P15). On the other hand, perceived prior knowledge in elements in the rubric made P14 perceive the rubric as rather "simple" with limited potential for learning and made P10 feel that there was "no need to probe deeper with GenAI". Moreover, participants' cognitive abilities to appropriate AI feedback also constrained learning, as P08 pointed out that GenAI feedback contained too much information that "did not directly address what [she] wanted," which made her at a loss in how to proceed with revision ("After reading the suggestions, I still did not know how to revise"). This incapability made P10 choose to ask GenAI to revise for him, which lowered his cognitive engagement: "I didn't know how to revise. I used GenAI to revise many aspects, and I did not think much in the process".

Post hoc comparisons using Bonferroni correction (Table 3) further revealed that the two AI groups outperformed the No GenAI Involvement group in providing problem justification (Condition 1: $M = 2.82$, $SD = .25$; Condition 2: $M = 1.91$, $SD = .25$; Condition 3: $M = .77$, $SD = .25$), and in offering constructive revision suggestions (Condition 1: $M = 4.97$, $SD = .35$; Condition 2: $M = 4.18$, $SD = .36$; Condition 3: $M = 1.68$, $SD = .35$). Yet, findings on problem identification were mixed, where Condition 2 ($M = 1.85$, $SD = .35$) did not differ from Condition 3 ($M = 1.18$, $SD = .35$). The findings indicate that integrating GenAI into the feedback process contributed to writing skill development through boosting the abilities to give explanations and provide constructive advice on problems identified in essays, whereas its effect on the abilities of problem identification depended on how GenAI was integrated into the feedback process. Moreover, Condition 1 outperformed Condition 2 in problem identification ($M = 4.41$, $SD = .35$ vs $M = 1.18$, $SD = .35$) and in problem justification ($M = 2.82$, $SD = .25$ vs $M = 1.91$, $SD = .25$).

The interview responses further revealed that the observed edge in problem justification and constructive advice among the two AI groups might be due to interactivity with GenAI. Participants from the GenAI as Feedback Receiver group recounted that being able to interact with GenAI helped them understand unfamiliar concepts in the assessment rubric (P04; P05). P02 further shared that "asking GenAI about its thinking process made [her] understand how to write rebuttal". Similarly, participants from the GenAI as Feedback Provider group shared that GenAI enabled them to access "explanations on confusing concepts" (P06) and "seek examples on how to address the weak points identified by GenAI" (P07). Thus, interaction with GenAI gave these two groups exposure to justification and revision advice. Yet, the GenAI as Feedback Receiver group had higher instances of learning-oriented interaction with AI, which accounts for their greater performance in problem justification.

The mixed findings on problem identification resulted from the interplay of engagement in evaluation (or lack thereof) with GenAI support (or lack thereof). Using the assessment rubric to evaluate essays made the participants "notice aspects [they] didn't pay attention to before" (P04, P11). Yet, not having external support from AI, the majority of the participants from the No GenAI Involvement condition found themself unable to identify specific problems: "I'm not clear about some points in the rubric framework, which made me unable to have a clear idea on the problems in my essay" (P13); and "I could identify problems on big issues, such as the lack of a rebuttal or evidence to support a claim. But for the other aspects, I couldn't identify my problems" (P14). This was exacerbated by the lack of opportunities to interact with GenAI for elaboration or guidance, which made them unable to see the need

Table 3
Differences in essay evaluation ability.

		Mean (SD)	Pairwise comparison
Problem identification score	Condition 1: GenAI as Feedback Receiver (n = 34)	4.41 (.35)	1 > 2***
			1 > 3***
	Condition 2: GenAI as Feedback Provider (n = 33)	1.18 (.35)	2 < 3
	Condition 3: No GenAI Involvement (n = 34)	1.85 (.35)	
Problem justification score	Condition 1: GenAI as Feedback Receiver (n = 34)	2.82 (.25)	1 > 2*
			1 > 3***
	Condition 2: GenAI as Feedback Provider (n = 33)	1.91 (.25)	2 > 3***
	Condition 3: No GenAI Involvement (n = 34)	.77 (.25)	
Constructive advice score	Condition 1: GenAI as Feedback Receiver (n = 34)	4.97 (.35)	1 > 2
			1 > 3***
	Condition 2: GenAI as Feedback Provider (n = 33)	4.18 (.36)	2 > 3***
	Condition 3: No GenAI Involvement (n = 34)	1.68 (.35)	

Note. ***$p < .001$; *$p < .05$.

for revision. For instance, P11 commented: "I need someone to tell me whether and why a certain aspect of my essay needs further improvement". Similarly, P15 expressed the wish for "a comprehensive analysis [of the rubric] so that [he] wouldn't feel lost in why and how to improve". For the GenAI as Feedback Provider group, as the evaluation work was offloaded to AI, who helped "identify points [they] could not spot on [their] own" (P07), they didn't get a chance to directly practice problem identification, which explained why they performed the worst in this category.

4.2.3. Differences in writing self-efficacy

ANCOVA analysis on writing self-efficacy (Table 4) revealed a significant difference across the three conditions ($F(2, 97) = 5.40$, $p < .01$, partial $\eta^2 = .10$), with significantly higher gains for Condition 1 ($M = 4.74$, $SD = .55$) and no significant differences between Condition 2 ($M = 4.42$, $SD = .53$) and Condition 3 ($M = 4.45$, $SD = .78$).

Although all three groups reported boosted writing self-efficacy, participants from the GenAI as Feedback Receiver condition attributed it to the acquired metacognitive knowledge of argumentative writing and of AI. For some, seeing different writing techniques in GenAI-generated model texts (P01 and P03) and retrieving elaborative examples on specific criteria from GenAI (P04) equipped them with metacognitive understanding that could guide future writing. For others, "being able to improve" GenAI-generated essays made P01 and P02 believe in their abilities to write good essays. Perceiving that AI essays, though sounded elegant, read like "piecing data together," P02 believed that "[she] may be able to write more lively essays, more humane essays".

Participants from the GenAI as Feedback Provider condition reported rather divided views. On the one hand, the acquired metacognitive knowledge of argumentative writing "eased the writing task" and could "be transferred to new writing tasks" (P06), consequently making them "feel more confident in writing a good argumentative essay" (P09). Receiving "positive remarks in GenAI feedback" also boosted self-efficacy (P07). On the other hand, performance-oriented engagement with AI for revision made them uncertain about their progress, giving them "a false sense of confidence" (P09; P10). P09 elaborated: "AI helped me make a better revision, but I might have relied on it too much that my teacher wouldn't be able to see my real level when reading my revised essays".

Similar mixed feelings were expressed by participants in the No GenAI Involvement group. Acquiring organizational structure from the rubric gave them a sense of certainty (e.g., "have a better idea on where and how to start, no longer feeling at a lost" – P15; "a guarantee on meeting basic standards" – P13) and a sense of achievement (e.g., "seeing more and more boxes on the self-checklist being ticked, I felt motivated and satisfied" – P11). They consequently felt "freer" (P12 and P15) and "bolder" (P13) in writing. But at the same time, they also expressed a sense of uncertainty about the metacognitive knowledge (e.g., "I feel my hook is attractive, but may not be perceived so by others" – P11; "I'm not sure whether I would be able to apply the framework during exam when the rubric is no longer available" – P13). P11 also pointed out that because the assessment rubric was fixed across the four writing tasks, she spotted fewer and fewer problems, which might have given her a false sense of self-efficacy as "there might still be many problems which simply were not covered in the rubric". This false sense of self-efficacy reduced P12's cognitive engagement with the rubric, where she reported "skimming through" and "paying attention to only a few elements" in later rounds.

5. Discussion

Driven by the observed decoupling of performance gains and writing development and the concerns over potential cognitive atrophy in AI-assisted writing (Fan et al., 2025; Henderson et al., 2025; Wang & Wang, 2025), this study explored ways to optimize human-AI interaction that supports writing development from the angle of task offloading. It revealed that: (1) offloading either the drafting or the evaluation task to GenAI led to greater writing development than no offloading; and (2) offloading the drafting task to GenAI was most effective. Joining others (Darvishi et al., 2024; Li & Tekwa, 2025), the study underscores the significance of task design in boosting the learning potentials of human-AI interaction. Existing discourses on task design emphasize scaffolding – gradually shifting externalized support from GenAI towards internal monitoring (Darvishi et al., 2024), and timing – involving GenAI only after students have completed an initial draft independently (Kosmyna et al., 2025; Li & Tekwa, 2025). This study drew attention to yet another factor – the nature of task offloading, advocating offloading the generation tasks (e.g., drafting essays) to AI and onloading the metacognitive tasks (e.g., evaluation) to learners as a way to counter potential cognitive atrophy in AI-assisted writing. This task offloading condition was found to yield the greatest generative learning potential, adding to the nascent evidence on the developmental advantages of critiquing AI-generated texts (Hong et al., 2025; Li, R., 2025).

More importantly, this study revealed psychological mechanisms behind the greater learning potential of this task offloading condition – deeper cognitive engagement, greater metacognitive awareness of argumentative writing and AI, and boosted writing self-

Table 4
Differences in the change of writing self-efficacy.

	Pre-writing efficacy Mean (SD)	Post-writing efficacy Mean (SD)	Pairwise comparison
Condition 1: GenAI as Feedback Receiver (n = 34)	3.61 (.59)	4.74 (.55)	1 > 2 * 1 > 3 *
Condition 2: GenAI as Feedback Provider (n = 33)	3.68 (.72)	4.42 (.53)	3 > 2
Condition 3: No GenAI Involvement (n = 34)	3.79 (.53)	4.45 (.78)	

Note. *$p < .05$.

efficacy. Deeper cognitive engagement derived from the high-quality activation of the visualization mode conceptualized in the Generative Sense Making framework (Fiorella, 2023). The high-quality activation manifested in deeper cognitive engagement with the visualization tool (the assessment rubric) due to heightened interaction with GenAI, learning-oriented interaction in particular, around the rubric; and in additional concrete, enriched, and elaborated visualization resources (AI-generated model texts) accessible in this condition, which enabled attention to higher-order aspects of writing during essay evaluation and revision. In contrast, the visualization mode was rather limiting for the other two conditions due to either lacking opportunity or incentive to interact with GenAI or being stuck dealing with lower-order issues, such as grammar and word choice, in their own drafts. Offloading the generation task to GenAI while onloading the metacognitive task to learners lowers the intrinsic cognitive load of writing (i.e., the mental efforts in lexical retrieval, grammatical encoding, and idea formulation) but maintains the germane cognitive load (i.e., the mental efforts dedicated to constructing and automating schemas), which, according to Cognitive Load Theory (Sweller, 2011), may induce thorough and deep engagement with the learning task without feeling overwhelmed (e.g., Evans et al., 2024). Engaging in the metacognitive task of providing feedback activates in-depth engagement with the content and comparison with alternatives, elevating reflexivity (Nicol, 2012; Zong et al., 2021).

High-quality activation of the visualization mode contributes to the activation of the explanation mode, i.e., the formulation of underlying principles (Fiorella, 2023). In this study, the elevated cognitive engagement with the assessment rubric was found to induce deeper metacognitive knowledge of argumentative essay that was transferred to the peer review task. This finding makes sense, since engagement with rubrics and model-text analysis are associated with the construction of internal schema and deeper metacognitive awareness of writing (Zeng et al., 2023; Zhang et al., 2024). Evaluating and revising AI essays led to metacognitive awareness of AI's limitations in writing, which further augmented this group's metacognitive knowledge of argumentative essay. This finding aligns with previous studies showing that critiquing and analyzing AI-generated compositions may enhance metalinguistic awareness of writing and metacognitive understanding of genre structure (Chambers & Owen, 2024; Li, R., 2025). This depth of metacognitive knowledge was neither salient among the GenAI as Feedback Provider group whose task only involved the enacting mode (essay revision), nor achievable among the No GenAI Involvement group who had no access to AI, leading to only fuzzy, surface-level metacognitive knowledge that served as a reminder or a general direction at most.

The acquired metacognitive knowledge on both argumentative essay and AI's limitations in writing further fueled this group's writing self-efficacy, in contrast to the false sense of self-efficacy among the other two groups as a result of perceived reliance on AI or perceived ambiguity in metacognitive knowledge. Previous research has suggested the link between AI literacy, critical evaluation of AI technologies in particular, and writing self-efficacy (e.g., Pan & Zhang, 2025). This study further revealed that being able to pinpoint and amend problems of the more capable peer that they regarded GenAI to be activated a strong sense of affirmation, freeing the participants from the social pressure and obstacles often associated with peer feedback (Martin & Sippel, 2021; Zhou et al., 2020).

Taken together, the findings suggest a framework of psychological mechanisms underlying the learning potential of offloading drafting to AI and onloading evaluation and revision to learners (Fig. 3). The framework underscores three roles of GenAI in the process: (1) as an interaction medium that enhances the level of cognitive engagement and deepens metacognitive knowledge of argumentative essay; (2) as a draft provider that channels cognitive attention to higher-order aspects and raises metacognitive awareness of AI's limitations in writing; and (3) as a feedback receiver that provides a source of affirmation for self-efficacy. This process is shaped by an array of barriers identified through interview data in this study, including motivational barriers – perceived

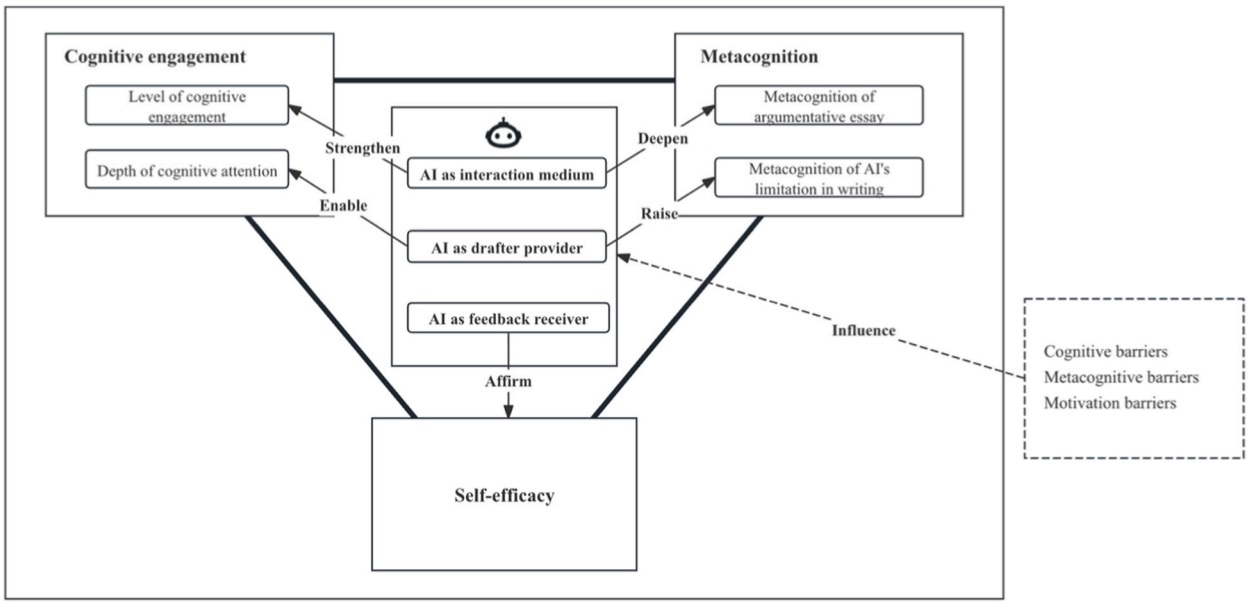

Fig. 3. Psychological mechanisms behind the generative learning potentials of GenAI as Feedback Receiver task design and AI's roles.

limitations of the assessment rubric and of GenAI output; metacognitive barriers – metacognitive attention, metacognitive positioning of AI, and strategic knowledge of AI interaction; and cognitive barriers – language proficiency, prior knowledge of the rubric, and capacities to appropriate AI output.

The findings contribute to the scholarly discussions on cognitive offloading (the use of external tools to reduce internal cognitive demands) in human-AI interaction. Jose et al. (2025) differentiated *assistive offloading* (i.e., offloading that assists cognition) from *substitutive offloading* (i.e., offloading that replaces cognition through auto-suggestion), and argued that the latter may induce shallow engagement, illusion of competence, reduced agency, and encoding and retrieval decline. Findings from this study support the conceptualized psychological differences induced by the level of cognitive offloading. Offloading drafting to AI and onloading evaluation to the learner positions GenAI as a tool to assist the productive struggles in evaluation, which are associated with greater engagement with the content, greater internalization of learning, and more reflexivity towards one's ideas (Kapur, 2016; Pak, 2024). Deeper cognitive engagement and critical reasoning, and greater metacognitive knowledge and reflexivity are conditions essential to realizing the benefits of GenAI in writing development (Zaheer et al., 2025). Adding to existing discourse, this study further revealed that substitutive offloading of the generation task of drafting, when supplemented with critiquing and revision, was less detrimental to learning than substitutive offloading of the metacognitive task of evaluation. The finding hence cautions that offloading essay evaluation to AI for automatic feedback, a common practice in AI-assisted writing, might be suboptimal for learning. Thus, cognitive offloading in human-AI interaction needs to consider not only the level of offloading but also the nature of tasks being offloaded.

6. Implications

The study has the following theoretical implications: First, the study adds nuances to cognitive offloading in human-AI interaction. Though cognitive offloading reduces cognitive load and facilitates engagement and performance, existing literature cautions that excessive or premature offloading may harm learning (Hong et al., 2025; Jose et al., 2025; Kosmyna et al., 2025), underscoring the level and timing of offloading. This study draws attention to yet another dimension of cognitive offloading – the type of task being offloaded, advocating future research to examine the effects of task offloading in different learning contexts and among different student populations. Second, the study conceptualizes three roles of GenAI in amplifying learning potentials of generative learning tasks that activate visualization and enacting modes (Fiorella, 2023), illustrating how it might empower cognitive, metacognitive, and affective processes. Future research may examine these roles separately to understand the amplification potential of each. Moreover, the extent to which each psychological mechanism contributes to writing development and how these mechanisms might interplay remain unknown and deserve further research. Future research may also examine how the working mechanisms might differ when learners provide feedback to GenAI versus to peers.

The study has the following practical implications: First, the study advises caution regarding the widespread use of GenAI as feedback provider, and urges exploration into alternative task-offloading approaches, such as using GenAI as writing text providers (Grobe, 2023; Li, 2025), especially when the pedagogical purpose is to develop metacognitive awareness of writing. Second, the study draws attention to an array of motivation, cognitive, and metacognitive barriers that may undermine the learning potential of GenAI-assisted writing. Teachers hence need to create conditions in pedagogical design and learner interventions to help learners overcome these barriers. In addition to enhancing prior knowledge, reducing the cognitive load in the task design, or providing explicit instruction, scaffolded practice, and independent practice to support learners' internal sense-making processes as recommended by Fiorella (2023), this study suggests the following instructional practices: (1) to design personally meaningful writing tasks that incentivize learning-oriented interaction with GenAI; (2) to arrange collaborative work around GenAI feedback to enhance revision skills; and (3) to implement learner training that helps learners build collaborative relationships with GenAI, position AI as a learning guide and learning partner, and engage in proactive and reflective use of GenAI feedback in writing (Yang et al., 2024; Zaheer et al., 2025; Zhan et al., 2025).

7. Conclusions, limitations, and future research

Adopting a quasi-experimental design, this study examined the effects of task offloading approaches on writing development. It found that involving GenAI in the feedback loop facilitated learning and that offloading drafting to GenAI and onloading evaluation and revision to learners carried the most benefit. The study further revealed higher cognitive engagement, deeper metacognition about argumentative writing and AI, and boosted self-efficacy under this task offloading condition. The findings advocate greater attention to task offloading in human-AI interaction.

This study has several limitations that call for future research. First, this study was conducted in the argumentative writing context. As students' interaction with GenAI and abilities to reap the benefits may vary for different learning tasks and different types of writing genres (Liu et al., 2024; Vaccaro et al., 2024), the observed edge of positioning GenAI as draft producers for learners to evaluate and revise needs to be investigated in different learning and writing contexts.

Second, several sample-related limitations should be acknowledged. The small sample size of 101 participants across three experimental conditions may have limited statistical power and increased the sensitivity of the results to sampling variability across groups. Although we controlled for prior writing ability and self-efficacy, the groups may vary in other influencing factors such as task motivation and AI prompt literacy. Future studies should employ a larger participant pool to verify the robustness of these findings. Furthermore, the participants were sampled from a single university, predominantly female, and of intermediate-high English proficiency. The demographic and institutional homogeneity may constrain the external validity of the results. Learners of different language proficiency levels and gender benefit differently from feedback providing (Cao et al., 2019; Lundstrom & Baker, 2009; Zeng

& Ravindran, 2024), and language and prior knowledge are important cognitive barriers that affect learning potential. Future research may replicate this study with diverse populations to further validate the preliminary observations from this study.

Third, the research tools are limiting. The static assessment rubrics used in the intervention might have a diminishing impact on learning over time, especially among the No GenAI Involvement condition, as revealed in interview responses. Future studies may repeat the study with dynamic rubrics. Using a single GenAI tool, Kimi, with its unique algorithmic biases or functionalities and low AI explainability (i.e., invisible chain of thought) at the time of the study might have constrained the power of AI and shaped AI interaction behaviors. The findings hence need to be validated using more updated AI tools and allowing coordinated use of multiple AI tools. This study used peer review performance as a behavioral proxy of metacognitive knowledge of argumentative writing. Although peer review indexes domain knowledge – the groundwork for metacognitive knowledge – and could reasonably serve as a window into metacognitive knowledge, it is a partial indicator at most and does not directly tap into metacognitive awareness of the argumentative writing process. Future research may complement peer review measures with additional methods (e.g., self-report inventories and think-alouds) on metacognitive reasoning to obtain a more complete picture.

Fourth, the research design limits the findings. While the seven-week intervention, which included four writing tasks, contained sufficient intensity to observe immediate effects, the short duration could not speak to sustained writing skill development, metacognitive development, and self-efficacy, as writing development, particularly at the argumentative level, requires extended practice across diverse genres and contexts. Thus, the findings are preliminary at most. We recommended further research to employ a longer intervention design to shed light on the evolving impact of task offloading conditions on writing skill development and psychological processes across different types of writing tasks, genres, and contexts. Also, for the No GenAI Involvement condition, although we implemented measures to prevent unauthorized GenAI usage, the prevalence of GenAI tools made absolute control impossible. Future investigations may consider conducting the interventions in controlled environments, such as in class, to effectively monitor and control for external GenAI usage. This study, although based on a mixture of behavioral and self-report data sources, relied primarily on retrospective measures to shed light on psychological processes, which might not have fully captured these processes and could not shed nuanced insights into cognitive processing and AI interaction. Future studies should consider incorporating process-tracing measures to capture these processes in real time and log moment-by-moment AI interaction to conduct detailed analysis into psychological processes and interaction dynamics that underpin the learning gains. Moreover, due to the sample constraints, the study was only able to create three conditions to compare the effects of cognitive offloading on drafting and on evaluation. To shed a comprehensive light on cognitive offloading in writing feedback loop, future studies may add a fourth condition – cognitive offloading on revising (learner draft, AI revise, learner evaluate the revision). Including this condition in the design would enable us to derive nuanced insights into the respective effects of offloading the two generation tasks in writing (draft vs. revision).

Finally, it should be noted that the influencing factors (e.g., motivational barriers, metacognitive barriers, and cognitive barriers) in Fig. 3 were derived from the interview responses. Future studies are therefore needed to test their potential associations with the task offloading effect through measuring them and including them in statistical analyses. Furthermore, the study focused on understanding the aggregated effects of task offloading conditions. However, individual differences in AI literacy, prompt-engineering skills, and prior experience with GenAI tools might have influenced the outcomes and could serve as potential confounding variables. Future research should examine how these individual differences moderate the effects of task offloading approaches on writing development, and discern differential learning trajectories or identify specific subgroups that might benefit more or less from distinct GenAI task loading approaches.

CRediT authorship contribution statement

Chun Lai: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mengru Pan:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Kai Guo:** Writing – review & editing, Visualization, Validation, Formal analysis, Data curation. **Yaqiong Cui:** Writing – review & editing, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve readability and language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Declaration of competing interest

The authors report there are no competing interests to declare.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compedu.2026.105675.

Appendix A. Demographic information of participants

Class	Age	Gender	English proficiency (CET-4 scores)	The frequency of GenAI use [a]
Class 1 (N = 34)	M = 20.18	Male = 3; Female = 31	M = 478.03; SD = 36.61	3.06
Class 2 (N = 33)	M = 20.55	Male = 9; Female = 24	M = 495.63; SD = 36.62	3.12
Class 3 (N = 34)	M = 20.44	Male = 3; Female = 31	M = 482.5; SD = 46.69	3.32

Note. a. 1 = a few times per year; 2 = a few times per month; 3 = a few times per week; 4 = less than 30 min per day; 5 = less than 1 h per day; 6 = 1 h to 3 h per day; 7 = more than 3 h per day.

Appendix B. The argumentative writing evaluation rubric

Aspects	4 points - Excellent	3 points - Good	2 points - Fair	1 point - Poor
Introduction *Original hook engages the reader's curiosity; background information is relevant and clearly develops into thesis; thesis/claim is strong, clearly stated.*	Engaging hook captures the reader's curiosity; background information is relevant and leads to a strong, clear thesis.	Hook is somehow engaging; background information is relevant and leads to a clear thesis.	Hook is weak; background information is vague or not clearly connected to the thesis; thesis is unclear or weak.	No hook; background information is irrelevant or missing; thesis is absent or poorly stated.
Reason & Evidence *Two or more main points are well developed with supporting details, and all the evidence is relevant and clearly connected to the thesis.*	Two or more main points are well developed with relevant supporting details; all evidence is clearly connected to the thesis.	Main points are developed with supporting details; evidence is mostly relevant and connected to the thesis.	Main points are underdeveloped or lack supporting details; evidence is somewhat relevant but lacks clear connection to the thesis.	Main points are missing or unclear; evidence is irrelevant or absent.
Counterargument & Refutation *Counterclaim is effectively addressed; strong rebuttal is supported by compelling evidence and effectively refutes counterclaim.*	Counterclaim is effectively addressed; strong rebuttal is supported by compelling evidence that thoroughly refutes the counterclaim.	Counterclaim is addressed; rebuttal is present and supported by some evidence but may not fully refute the counterclaim.	Counterclaim is weakly addressed; rebuttal is present but lacks sufficient evidence or clarity.	Counterclaim is not addressed or absent; rebuttal is missing or ineffective.
Conclusion *Conclusion summarizes the main topics without repeating previous sentences; writer's opinions and suggestions for change are logical and well thought out.*	Conclusion effectively summarizes main topics without repetition; writer's opinions and suggestions are logical and well thought out.	Conclusion summarizes main topics; opinions and suggestions are logical but may lack depth.	Conclusion is vague or repeats previous content; opinions and suggestions are unclear or poorly developed.	Conclusion is missing or does not summarize main topics; no logical opinions or suggestions provided.
Organization & Transition *Ideas are coherently organized into separate body paragraphs; topic sentences are complete; logical, compelling progression of ideas in essay; effective, mature, graceful transitions exist throughout the essay.*	Ideas are coherently organized into separate body paragraphs; topic sentences are complete; logical progression and effective transitions enhance the essay.	Ideas are mostly organized; topic sentences are present; progression is clear with generally effective transitions.	Organization is unclear; topic sentences are incomplete; progression of ideas is weak; transitions are lacking.	Ideas are chaotic; topic sentences are absent; no logical progression or transitions present.
Language Use *Tone is formal; no personal pronouns; no slang; strong words appropriate to stance are used; no repetition of major words; punctuation, capitalization, spelling and usage are correct.*	Tone is formal with no personal pronouns or slang; strong, varied word choice enhances the argument; conventions (punctuation, capitalization, spelling) are correct.	Tone is mostly formal; word choice is appropriate; few errors in conventions that do not interfere with understanding.	Tone is somewhat formal; word choice is basic or repetitive; several errors in conventions that may hinder understanding.	Tone is informal; inappropriate word choice; frequent errors in conventions that significantly hinder understanding.

Appendix C. The evaluation worksheet for Conditions 1 and 3

		Please evaluate the essay against the checklist (put a ✔ on the items that are in place)	Areas for improvement
Introduction	Original hook engages the reader's curiosity	☐ 1. Is there a hook in the introduction paragraph? ☐ 2. Is the hook successful in arousing your interest in reading?	
	Thesis/claim is strong, clearly stated, and supported by relevant background information	☐ 1. Did the author provide the background information that leads to the thesis? ☐ 2. Is the thesis/claim clearly stated? ☐ 3. Does the thesis state a significant and compelling position?	
Organization & Transition	Two or more main points are well developed with relevant supporting details clearly connected to the thesis	☐ 1. Is each main point well supported by evidence? ☐ 2. Is the evidence relevant to each main point? ☐ 3. Did the author use different types of evidence to support the main points?	
	Ideas are sequenced logically and coherently organized into separate body paragraphs with clear topic sentences	☐ 1. Does each paragraph center around a main idea/ point? ☐ 2. Does each paragraph start with a compelling topic sentence for each idea/point? ☐ 3. Is the sequence of the main points logical and shows a clear progression in deepening the argumentation? ☐ 4. Are the paragraph(s) well connected with one another with transition words?	
Counterclaim & Rebuttal	Strong rebuttal is supported by compelling evidence and effectively refutes counter claim	☐ 1. Is there a rebuttal to counter the main thesis/claim? ☐ 2. Are there relevant evidences to support the rebuttal ☐ 3. Is the rebuttal effective in refuting the main thesis/ claim?	
Conclusion	A strong conclusion with summary of the main points and writer's opinions and suggestions for change	☐ 1. Did the conclusion effectively summarize the main topics and thesis without repeating previous sentences? ☐ 2. Did the conclusion go beyond restating the thesis and includes a call to action?	
Language Use	Language is formal, with strong words and no errors	☐ 1. Are the tones formal and consistent without personal pronouns and slang? ☐ 2. Are words appropriate and strong to the stance and not too repetitive? ☐ 3. Are sentences strong and expressive with varied structure? ☐ 4. Is the essay free of punctuation, capitalization, spelling and grammar errors?	

Appendix D. The writing prompts used in the pre- and post-tests and during the intervention

	Task prompts
Pre-test	"Young people should be encouraged to pursue long-term, realistic goals rather than seek immediate fame and recognition." Discuss the extent to which you agree or disagree with this statement and provide reasons for the position you take. In developing and supporting your position, you should include relevant evidence. Write 300-500 words.
Intervention	'The best way to teach – whether as an educator, employer, or parent – is to praise positive actions and ignore negative ones.' Discuss the extent to which you agree or disagree with this statement and provide reasons for the position you take. In developing and supporting your position, you should include relevant evidence. Write 300-500 words. 'Governments should offer college and university education free of charge to all students.' Discuss the extent to which you agree or disagree with this statement and provide reasons for the position you take. In developing and supporting your position, you should include relevant evidence. Write 300-500 words. 'The luxuries and conveniences of contemporary life prevent people from developing into truly strong and independent individuals.' Discuss the extent to which you agree or disagree with this statement and provide reasons for the position you take. In developing and supporting your position, you should include relevant evidence. Write 300-500 words. 'College students should be encouraged to pursue subjects that interest them rather than the courses that seem most likely to lead to jobs.' Discuss the extent to which you agree or disagree with this statement and provide reasons for the position you take. In developing and supporting your position, you should include relevant evidence. Write 300-500 words.
Post-test	'Students should always question what they are taught instead of accepting it passively.' Discuss the extent to which you agree or disagree with this statement and provide reasons for the position you take. In developing and supporting your position, you should include relevant evidence. Write 300-500 words.

Appendix EWriting self-efficacy questionnaire

1. I can present well-rounded perspectives.
2. I can support my claims with effective evidence.

3. I can present logical argumentation.
4. I can clearly express my viewpoints.
5. I can use precise language to make effective arguments.
6. I can use persuasive language to articulate my arguments.
7. I can effectively use language to convey ideas with clarity and logic.

Appendix FPeer essay used in the peer review task

In the education system, many students are encouraged to listen to their superiors and rarely question what they say. However, it is my belief that students should sometimes question what they are taught, especially if they feel like their teacher is incorrect.

To begin, if students just accept every concept they are taught throughout school, they will lose the ability to think freely for themselves. For example, in my Literature class, my teacher encouraged us to speak up and give our opinions when discussing a piece of literature. Each novel or poem we read, we always analyzed and discussed them in class. If my teacher had just told us about his opinions on what the meaning of each novel was, and our class never gave our input, we would lose valuable thinking abilities. By actively imputing our opinions, we gradually learned how to become more independent thinkers.

However, there are some cases when students should not question what they are taught. A science class is an example of a situation where students should limit what they question. Many concepts in science, such as the law of gravity, is static and will never change. Students need to learn when it is an appropriate time to challenge what they are taught and when to accept what they are taught.

On the basis of the points mentioned above, I neither agree nor disagree with the statement that students should always question what they are taught instead of accepting it passively.

Appendix G. Interview protocol for students in the three conditions

Questions for students in Condition 1	• What's your overall perception of the process of evaluating and revising the GenAI-generated essay? • Did you find this process helpful in enhancing your argumentative writing? If yes, what aspects and how? And on what aspects you didn't find it helpful and why not? • How did you find this arrangement in shaping your writing self-efficacy and writing development? In what aspects was this arrangement helpful? In what aspects was it not helpful? • How did you perceive the role of GenAI in writing?
Questions for students in Condition 2	• What's your overall perception of the process of seeking feedback from GenAI and revising your essay based on the feedback? • Did you find this process helpful in enhancing your argumentative writing? If yes, what aspects and how? And on what aspects you didn't find it helpful and why not? • How did you find this arrangement in shaping your writing self-efficacy and writing development? In what aspects was this arrangement helpful? In what aspects was it not helpful? • How did you perceive the role of GenAI in writing?
Questions for students in Condition 3	• What's your overall perception of the process of evaluating and revising your essay based on the self-checklist? • Did you find this process helpful in enhancing your argumentative writing? If yes, what aspects and how? And on what aspects you didn't find it helpful and why not? • How did you find this arrangement in shaping your writing self-efficacy and writing development? In what aspects was this arrangement helpful? In what aspects was it not helpful?

Appendix H. Coding framework for analyzing conversation logs

Category	Description	Examples	Condition 1 (N)				Condition 2 (N)			
			Task 1	Task 2	Task 3	Task 4	Task 1	Task 2	Task 3	Task 4
Prompt amount	The total number of prompts a student sends to GenAI.	/	217	250	227	212	63	81	80	83
Learning-oriented prompts	Prompts that ask GenAI for suggestions, guidance, feedback, explanations, information, or elaborations that create opportunities for learning.	*What is a hook in an introduction paragraph? Please give me an example of a hook?*	160	162	129	105	61	78	68	67

Appendix I. Coding framework for analyzing quality of peer feedback

Feature of feedback	Points	Label	Description
Identification	0	Poor - not mentioned	The comment did not include explicit identification of problem
	1	Average - mentioned but not localized	The comment included identification of problem without localization of identified problem
	2	Good - mentioned and localized	The comment included explicit and localized identification of problem
Justification	0	Poor - not mentioned	The comment did not include elaborations and justifications of identified problem
	1	Average - mentioned elaborated, but not justified	The comment included elaborations but not justifications of identified problem
	2	Good - mentioned elaborated and justified	The comment included elaborations and justifications of identified problem
Constructive	0	Poor - not mentioned	The comment did not include any recommendations or action plans for further improvements.
	1	Average - only recommendation is mentioned	The comment included recommendations but not action plans for further improvements.
	2	Good - both recommendation and action plan are mentioned	The comment included recommendations and action plans for further improvements.

Appendix J. Coding scheme of the psychological processes and barriers

Themes	Categories	Sub-categories	Descriptions	Condition 1		Condition 2		Condition 3	
				Counts (N)	Representative quotes	Counts (N)	Representative quotes	Count (N)	Representative quotes
Cognitive engagement	Depth of cognitive processes attended to	Attention to and deep engagement with AI output	Carefully read and process AI-generated text or feedback in detail and/or with a critical lens	5	*I took a more critical lens when reading its essays. You would carefully look for its mistakes.*	2	*I read the part AI revised and carefully savored it, and did feel its words were excellent.* *Judge whether agreed with the rationale AI provided.*	/	/
		Reflexivity towards one's essay	Engage in self-reflection of one's essay critically to achieve a better understanding of one's strength and weakness	5	*The rubric makes you see what is missing and engage in self-reflection.* *When assessing the points, you would think of your previous essays.*	0	/	5	*Look at the form and start to inspect whether I touched all the points.* *I notice aspects I didn't pay attention to before.*
		Shallow/ Uncertain engagement with the assessment rubric	Engage with the assessment rubric with uncertainty, lacking clarity	0	/	0	/	5	*When assessing my essay, I feel I'm 'wrestling' with the assessment rubric.* *I felt I wrote well, but then it did not look that well either.* *I hope it would give me more detailed instructions.*
	Level of cognitive engagement	Learning-oriented engagement AI output	Engage with AI-generated text or feedback to learn	4	*I study AI's essays to curate materials for writing.* *I would remember and use the vocabularies frequently appeared in its texts.*	1	*I study some new perspectives from AI feedback.*	/	/
		Learning-oriented	Converse with AI to seek for	5	*I would ask AI about its thinking*	2	*I often seek examples on how*	/	/

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Themes	Categories	Sub-categories	Descriptions	Condition 1		Condition 2		Condition 3	
				Counts (N)	Representative quotes	Counts (N)	Representative quotes	Count (N)	Representative quotes
		conversation with AI	guidance, advice, examples, etc. that would facilitate learning or engage iteratively with AI		*process. Referring to the rubric, I would evaluate AI's essays and ask it questions on different aspects.*		*to address the weak points identified by GenAI. I would question it iteratively.*		
		Performance-oriented conversation with AI	Converse with AI to ask AI to complete the task for them	2	*I ask AI to revise the essay based on my feedback.*	5	*I ask AI to directly revise the essay for me. I ask AI to edit my language.*	/	/
Metacognition	Metacognitive knowledge of argumentative writing	Awareness of rudimentary aspects of metacognitive knowledge	Awareness of the basic declarative knowledge of factors that influence one's writing performance (e.g., the key elements; the organizational structure)	5	*Argumentation flow is very clear.*	5	*I sometimes would only write the claim without evidences or only use one sentence to talk about the evidences. It reminds me to include a list of evidences.*	5	*Using hooks that draw my attention.*
		Awareness of higher-order aspects of metacognitive knowledge	Awareness of the procedural (knowledge of how to execute specific skills) and conditional (knowledge of the conditions to apply various cognitive actions) aspects of writing	5	*While revising, I would think about what style of argumentative essay I want. It forced me to think deeper about argumentative writing.*	3	*It showed me different ways to support my claims.*	1	*The rubric reminded me to notice what I didn't notice before.*
		Perceived depth of metacognitive knowledge	Negative evaluative judgement of the depth and transferability one's metacognitive knowledge	0	/	3	*AI helped me make a better revision, but I might have relied on it too much that my teacher wouldn't be able to see my real level when reading my revised essays.*	5	*I feel the hook is attractive, but may not be perceived so by others.*
			Positive/ affirmative evaluative judgement about transferring procedural/ conditional metacognitive knowledge	4	*I would remember how it varies sentence structures as well as the vocabularies it frequently uses and apply them in my next writing.*	2	*It helped us to think from different perspectives, enhancing critical thinking.*	1	*The rubric provides clues on where to start when I write my own essay.*
			Positive/ affirmative evaluative judgement about applying declarative/ rudimentary metacognitive knowledge	5	*The rubric provides ways to make the essay more comprehensive and logical.*	5	*That organizational framework made me write faster.*	5	*It gives a general direction for writing.*
	Metacognition of AI interaction	Metacognitive awareness of the limitations of AI	Awareness of the limitations of the feedback/advice from AI	1	*Its essays often followed the same pattern*	5	*It provided lengthy responses with too much information.*	/	/

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Themes	Categories	Sub-categories	Descriptions	Condition 1		Condition 2		Condition 3	
				Counts (N)	Representative quotes	Counts (N)	Representative quotes	Count (N)	Representative quotes
			Awareness of the limitations of conversing with AI	3	*It couldn't make revisions based on my feedback, which made me frustrated, feeling that it couldn't get my intention.*	4	*It couldn't produce the version I wanted.*	/	/
			Awareness of the weaknesses in AI-generated essays	5	*It generated very similar templates, repeating a list of frequently-used words.* *It reads like piecing data together.*	0	/	/	/
		Metacognitive awareness of the strength of AI	Awareness of the strength of AI interaction and output	5	*Its essays are quite elegant.* *The words it used were indeed excellent.*	5	*It could explain any concepts I'm not sure about.*	/	/
		Metacognitive positioning of AI	Regarding AI as a teacher/tutor or a learning partner during the process to support learning	5	*It's a learning partner who is at a higher level than me.*	1	*A strategic advisor. It helps me broaden perspectives.*	/	/
			Regarding AI as an assistant to provide direct help on writing or revision during the process to support performance	0	/	4	*It plays more of an assistant role. After I complete an essay, I ask it to point out weaknesses and ask it to revise.* *I am dominant, it assists. When I need assistance, I ask it to provide solutions.*	/	/
Self-efficacy	Positive self-efficacy		Perception of positive impact on their writing efficacy	5	*I may be able to write more lively essays, more humane, if I write seriously.*	3	*I can write more logical, engaging, and persuasive essays.*	2	*Seeing more and more boxes on the self-checklist being ticked in subsequent rounds, I felt motivated and satisfied.*
	Negative self-efficacy		Perception of no or negative impact on their writing efficacy	0	/	2	*It helped me complete the writing, which did not reflect my ability.*	3	*I'm not sure whether I would be able to apply the framework during exam when the rubric is no longer available.*
Barrier	Motivational barriers		Perceived limited value of the rubric and/or AI output/ conversation for learning	3	*It might not be that useful in helping me develop deeper understanding.*	4	*I feel some feedback from AI is not very useful.*	5	*I feel the rubric only raised issues without providing solutions.*
	Metacognitive barriers		Perceived hindrance of one's metacognitive attention, strategic knowledge of interacting with AI or learning	3	*I can't get AI to understand my intention.*	4	*I paid more attention to structure. When it indicated my tone was not strong enough, not formal, I wouldn't pay much*	3	*Each criterion is too general and abstract.*

(*continued on next page*)

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Themes	Categories	Sub-categories	Descriptions	Condition 1		Condition 2		Condition 3	
				Counts (N)	Representative quotes	Counts (N)	Representative quotes	Count (N)	Representative quotes
			habits in performing the task				*attention.* *It couldn't produce the version I wanted, maybe because my prompt was not clear.*		
	Cognitive barrier		Perceived lack of prior knowledge or language proficiency or capacity of using the rubric and AI output to perform the task	2	*I know what a hook is.*	3	*After reading the suggestions, I still did not know how to revise.*	5	*Sometimes, I feel bewildered.* *My vocabulary is very limited.*

Data availability

Data will be made available on request.

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